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DATE: 24/11/2024

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COURSE: INT 303 Assignment

LAB 1: Understanding Network Layers and TCP/IP Model using OWASP Broken Web

Application IP

Exercise 1: Understanding OSI and TCP/IP Model

Question:

List the OSI model layers and describe the function of each.

1. **Physical Layer:** This layer deals with the physical connection between devices, including cables, switches, and the transmission of raw bit streams over a network medium.
2. **Data Link Layer:** This layer is responsible for node-to-node data transfer and error detection/correction. It frames packets for the network layer and manages access to the physical medium.
3. **Network Layer:** It manages the routing of data packets across different networks. This layer is responsible for logical addressing, routing, and forwarding.
4. **Transport Layer:** Ensures reliable transmission of data segments between points on a network, providing error recovery and flow control. It manages segmentation and reassembly of data.
5. **Session Layer:** This layer establishes, manages, and terminates sessions between applications. It coordinates communication and maintains connections.
6. **Presentation Layer:** It translates the data format from application to network format. This layer is also responsible for data encryption and compression.
7. **Application Layer:** The topmost layer where end-user software applications interact with the network. It provides network services to applications and facilitates user-level functionalities.

Match each OSI layer with its corresponding TCP

TCP/IP	OSI
Application layer	Application layer Presentation layer Session layer
Transport layer	Transport layer
Internetwork layer	Network layer
Network access layer	Data link layer

Exercise 2: Ping the OWASP Broken Web Application

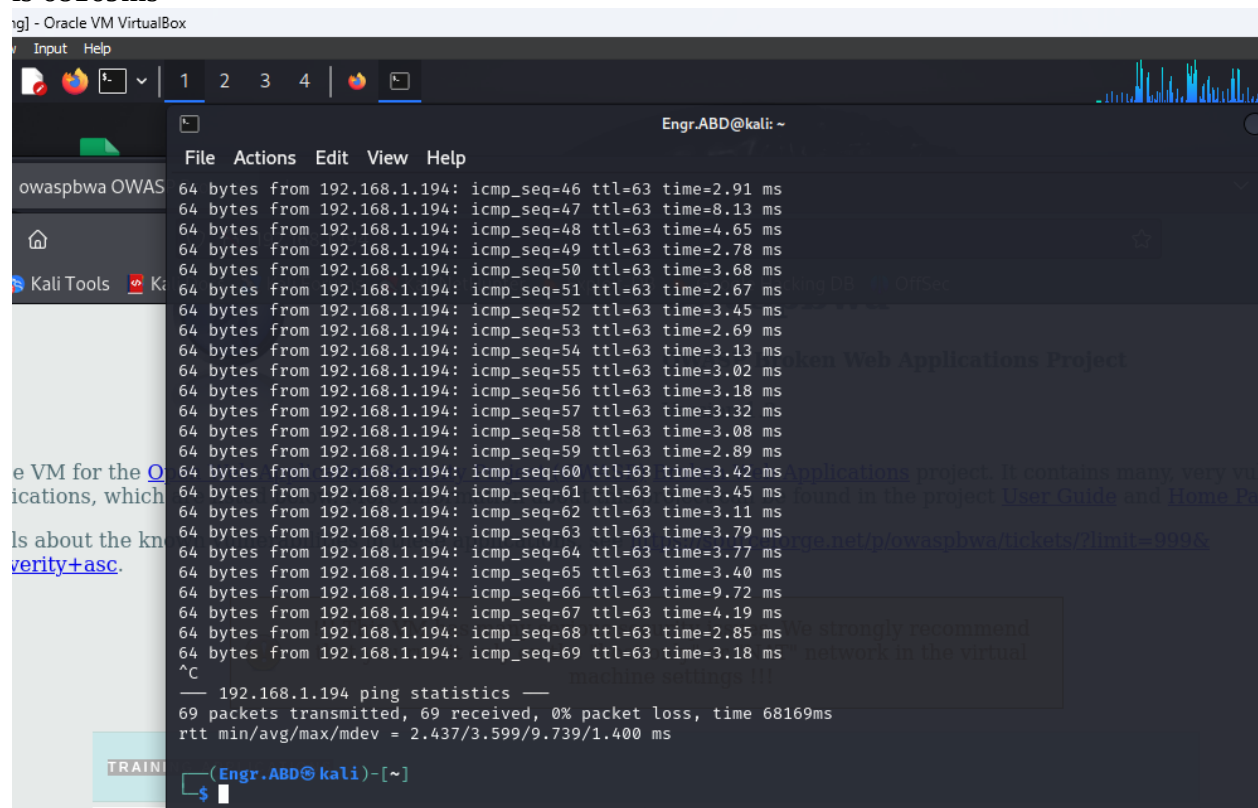
Task: Use the ping command to check the reachability of the OWASP Broken Web Application using its IP address. Command: ping <OWASP_IP> (replace <OWASP_IP> with the actual IP, e.g., ping 192.168.56.101)

Question:

What happens when you ping the OWASP application? Describe the process.

Transmitted packet data was observed within a short period of time, and also show that OWASP IP is reachable and available for communication, ICMP communication protocol is used while sent the data packet, this packet travel between my device and server go and back in milliseconds (time taken).

Ping statistics show that 69 packet transitted, 69 packet received, 0% loss, time for transmission is 68169ms



```
ngj - Oracle VM VirtualBox
v Input Help
1 2 3 4
File Actions Edit View Help
owaspbwa OWASP
Kali Tools
e VM for the O
ications, which
ls about the kn
erity+asc.
64 bytes from 192.168.1.194: icmp_seq=46 ttl=63 time=2.91 ms
64 bytes from 192.168.1.194: icmp_seq=47 ttl=63 time=8.13 ms
64 bytes from 192.168.1.194: icmp_seq=48 ttl=63 time=4.65 ms
64 bytes from 192.168.1.194: icmp_seq=49 ttl=63 time=2.78 ms
64 bytes from 192.168.1.194: icmp_seq=50 ttl=63 time=3.68 ms
64 bytes from 192.168.1.194: icmp_seq=51 ttl=63 time=2.67 ms
64 bytes from 192.168.1.194: icmp_seq=52 ttl=63 time=3.45 ms
64 bytes from 192.168.1.194: icmp_seq=53 ttl=63 time=2.69 ms
64 bytes from 192.168.1.194: icmp_seq=54 ttl=63 time=3.13 ms
64 bytes from 192.168.1.194: icmp_seq=55 ttl=63 time=3.02 ms
64 bytes from 192.168.1.194: icmp_seq=56 ttl=63 time=3.18 ms
64 bytes from 192.168.1.194: icmp_seq=57 ttl=63 time=3.32 ms
64 bytes from 192.168.1.194: icmp_seq=58 ttl=63 time=3.08 ms
64 bytes from 192.168.1.194: icmp_seq=59 ttl=63 time=2.89 ms
64 bytes from 192.168.1.194: icmp_seq=60 ttl=63 time=3.42 ms
64 bytes from 192.168.1.194: icmp_seq=61 ttl=63 time=3.45 ms
64 bytes from 192.168.1.194: icmp_seq=62 ttl=63 time=3.11 ms
64 bytes from 192.168.1.194: icmp_seq=63 ttl=63 time=3.79 ms
64 bytes from 192.168.1.194: icmp_seq=64 ttl=63 time=3.77 ms
64 bytes from 192.168.1.194: icmp_seq=65 ttl=63 time=3.40 ms
64 bytes from 192.168.1.194: icmp_seq=66 ttl=63 time=9.72 ms
64 bytes from 192.168.1.194: icmp_seq=67 ttl=63 time=4.19 ms
64 bytes from 192.168.1.194: icmp_seq=68 ttl=63 time=2.85 ms
64 bytes from 192.168.1.194: icmp_seq=69 ttl=63 time=3.18 ms
^C
— 192.168.1.194 ping statistics —
69 packets transmitted, 69 received, 0% packet loss, time 68169ms
rtt min/avg/max/mdev = 2.437/3.599/9.739/1.400 ms
(Engr.ABD@kali)-[~]
```

Which OSI layer does the ping command operate in? Explain.

The ping command operates at the Network layer (Layer 3) of the OSI model. It uses the Internet Control Message Protocol (ICMP) to send echo requests and echo reply to messages for network connectivity testing / to test connectivity and measure the round-trip time (RTT) between a computer and a destination host.

Exercise 3: Tracing the Path to the OWASP Application

Task: Use the traceroute command to trace the route packets take to reach the OWASP Broken Web Application.

Question:

How many hops did it take to reach the OWASP VM?

30 hops at maximum were to reach the OWASP_VM and 60 packets.

```
Engr.ABD@kali: ~  
File Actions Edit View Help  
  
(Engr.ABD@kali)-[~]  
$ traceroute 192.168.1.194  
traceroute to 192.168.1.194 (192.168.1.194), 30 hops max, 60 byte packets  
 1  10.0.2.2 (10.0.2.2)  58.438 ms  2.035 ms  1.675 ms  
 2  * * *  
 3  * * *  
 4  * * *  
 5  * * *  
 6  * * *  
 7  * * *  
 8  * * *  
 9  * * *  
10  * * *  
11  * * *  
12  * * *  
13  * * *  
14  * * *  
15  * * *  
16  * * *  
17  * * *  
18  * * *  
19  * * *  
20  * * *  
21  * * *  
22  * * *  
23  * * *  
24  * * *  
25  * * *  
26  * * *  
27  * * *
```

Describe the significance of each hop and what role traceroute plays in network troubleshooting

A hop in a traceroute is the point at which data moves from one router to the next. Traceroute is a command that network administrators use to troubleshoot network issues by identifying where latency or packet loss occurs. Here's what each hop and traceroute can tell you about network issues:

- **Hop number:** The sequence in which a router or device was encountered. The first hops are on your local area network (LAN), and subsequent hops are on routers controlled by your internet service provider (ISP).
- **Round-trip time (RTT):** The time it takes for a packet to travel from your device to a particular hop. RTT is measured in milliseconds. Higher RTTs can indicate network bottlenecks or congestion.

- **IP address or domain name:** Identifies the routers and systems involved in data transmission.
- **Hop patterns:** Sudden increases in a hop, or a pattern of increasing hops towards the destination, can indicate an issue.
- **Asterisks:** Asterisks (*) in the report indicate packet loss.

Exercise 4: Viewing Active Connections to OWASP VM

Task: Use netstat to view active network connections between your system and the OWASP Application

Question:

What connections do you see? Identify the source and destination IP addresses.

Transmission control protocol (tcp) was observed, the source, destination IP addresses and their port are 10.0.2.15:51398 and 192.168.92.135:80 respectively.

The screenshot shows a terminal window titled 'Engr.ABD@kali: ~'. The user has entered the command `netstat -an | grep 192.168.92.135`. The output shows a single active TCP connection from 10.0.2.15:51398 to 192.168.92.135:80 in the TIME_WAIT state. Below the terminal output, there is a banner for the Open Web Application Security Project (OWASP) Browser Security Project (BSP) and a warning box stating: '!!! This VM has many serious security issues that you run it only on the "host only" machine so you can't access the internet.'.

```

Engr.ABD@kali: ~
File Actions Edit View Help
(Engr.ABD@kali)-[~]
$ netstat -an | grep 192.168.92.135
tcp        0      0 10.0.2.15:51398    192.168.92.135:80  TIME_WAIT
(Engr.ABD@kali)-[~]
$

```

This is the VM for the [Open Web Application Security Project \(OWASP\) Browser Security Project \(BSP\)](#).
 web applications, which are listed below. More information about this project is available at [https://www.owasp.org/index.php/Browser_Security_Project](#).
 For details about the known vulnerabilities in these applications, see https://www.owasp.org/index.php/Browser_Security_Project sorted by [severity+asc](#).

!!! This VM has many serious security issues that you run it only on the "host only" machine so you can't access the internet.

Explain how the Transport Layer (TCP/UDP) is involved in this communication.

The Transport Layer (TCP/UDP) is involved in communication by breaking down data into smaller segments, managing communication sessions, and ensuring reliable data transmission:

Data breakdown

The Transport Layer breaks down data from the upper layers into smaller segments.

Communication sessions

The Transport Layer manages end-to-end communication sessions, including establishing and terminating IP communication sessions.

Data transmission

The Transport Layer ensures reliable data transmission through retransmission and a three-way handshake.

Error detection and correction

The Transport Layer provides error detection and correction.

Port numbers

The Transport Layer uses standardized port numbers to facilitate communication for well-known applications.

The two most widely used protocols that operate on the Transport Layer are Transmission Control Protocol (TCP) and User Datagram Protocol (UDP):

TCP: Provides a reliable data delivery service with end-of-end error detection and correction.

UDP: Provides low-overhead, connectionless datagram delivery service.

Exercise 5: TCP vs. UDP

Task: Investigate the difference between TCP and UDP by scanning the OWASP Broken Web Application.

Question:

What are the key differences between TCP and UDP in terms of reliability and speed?

TCP host is up at 0.045s while UDP 0.043s which means that UDP is faster in terms of speed than TCP and UDP is more reliable.

```
(Engr.ABD@kali)-[~]
$ nmap -sT 192.168.92.135
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-26 07:05 EST
Nmap scan report for 192.168.92.135
Host is up (0.045s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp    open  netbios-ssn
143/tcp    open  imap
443/tcp    open  https
445/tcp    open  microsoft-ds
5001/tcp   open  complex-link
8080/tcp   open  http-proxy
8081/tcp   open  blackice-icecap

Nmap done: 1 IP address (1 host up) scanned in 7.39 seconds
```

Based on the scan results, list which services on the OWASP application are using TCP and which are using UDP.

The service use by TCP are ssh, http, netbios-ssn, imap, https, Microsoft-ds, complex-link, http-proxy and blackice-icecap while UDP services is netbios-ns.

```
File Actions Edit View Help
(root@kali)-[/home/Engr.ABD]
# nmap -sU 192.168.92.135 -v

Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-26 07:52 EST
Initiating Ping Scan at 07:52
Scanning 192.168.92.135 [4 ports]
Completed Ping Scan at 07:52, 0.09s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 07:52
Completed Parallel DNS resolution of 1 host. at 07:52, 0.04s elapsed
Initiating UDP Scan at 07:52
Scanning 192.168.92.135 [1000 ports]
Discovered open port 137/udp on 192.168.92.135
Completed UDP Scan at 07:53, 12.37s elapsed (1000 total ports)
Nmap scan report for 192.168.92.135
Host is up (0.0043s latency).
Not shown: 999 open|filtered udp ports (no-response)
PORT      STATE SERVICE
137/udp    open  netbios-ns

Read data files from: /usr/bin/.. /share/nmap
Nmap done: 1 IP address (1 host up) scanned in 13.22 seconds
Raw packets sent: 2092 (98.704KB) | Rcvd: 7 (1.480KB)

(root@kali)-[/home/Engr.ABD]
#
```

Exercise 6: Discovering MAC Addresses with ARP

Task: Use the arp command to view the MAC address of the OWASP Broken Web Application.

Question:

What is the MAC address associated with OWASP VM's IP?

I don't find MAC address

```
File Actions Edit View Help
(Engr.ABD@kali)-[~]
$ arp -a | grep 192.168.92.135

(Engr.ABD@kali)-[~]
$ arp -ans
? (10.0.2.2) at 52:54:00:12:35:02 [ether] on eth0
192.168.92.135: icmp_seq=49 ttl=63 time=2.12 m
(Engr.ABD@kali)-[~]
$ arp -a 192.168.92.135
192.168.92.135: icmp_seq=50 ttl=63 time=2.15 m
192.168.92.135: icmp_seq=51 ttl=63 time=2.89 m
192.168.92.135: icmp_seq=52 ttl=63 time=2.37 m
192.168.92.135: icmp_seq=53 ttl=63 time=2.15 m
192.168.92.135: icmp_seq=54 ttl=63 time=1.92 m
192.168.92.135: icmp_seq=55 ttl=63 time=2.54 m
192.168.92.135: icmp_seq=56 ttl=63 time=2.20 m
192.168.92.135: icmp_seq=57 ttl=63 time=2.66 m
192.168.92.135: icmp_seq=58 ttl=63 time=2.11 m
192.168.92.135: icmp_seq=59 ttl=63 time=2.17 m
192.168.92.135: icmp_seq=60 ttl=63 time=1.93 m
192.168.92.135: icmp_seq=61 ttl=63 time=2.06 m
arp: in 1 entries no match found.
```

Explain the significance of ARP in the Data Link Layer and how it contributes to successful communication.

The Address Resolution Protocol (ARP) is a key part of the Data Link Layer that allows devices on a network to communicate by mapping IP addresses to MAC addresses.

When a device needs to send data to another device on the same network, it sends an ARP request to every device on the network asking for the destination's MAC address. The destination device then responds with its MAC address, which is stored in the ARP table.

Exercise 7: Capturing Network Traffic with Wireshark

Task: Use Wireshark or tshark to capture network packets between your machine and the OWASP Broken Web Application. Command: wireshark (to launch the GUI) or tshark -i host

Question:

Analyze the captured traffic. What protocols are in use?

The Protocol use is TCP and port 80

The image shows a Wireshark network traffic capture. The top pane displays a list of captured packets. The bottom pane shows the details of the selected packet (No. 17), which is a TCP segment.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.15	34.117.188.166	TLSv1.2	93	Application Data
2	0.002082555	34.117.188.166	10.0.2.15	TCP	60	443 → 34408 [ACK] Seq=
3	0.149833983	34.117.188.166	10.0.2.15	TLSv1.2	93	Application Data
4	0.149872847	10.0.2.15	34.117.188.166	TCP	54	34408 → 443 [ACK] Seq=
5	7.017650694	10.0.2.15	142.250.201.78	TLSv1.2	93	Application Data
6	7.020641067	142.250.201.78	10.0.2.15	TCP	60	443 → 39682 [ACK] Seq=
7	7.157823985	142.250.201.78	10.0.2.15	TLSv1.2	93	Application Data
8	7.157928376	10.0.2.15	142.250.201.78	TCP	54	39682 → 443 [ACK] Seq=
9	8.353089481	10.0.2.15	192.168.92.168	DNS	73	Standard query 0x503d
10	8.358383463	10.0.2.15	192.168.92.168	DNS	73	Standard query 0x0a3e
11	8.622611905	192.168.92.168	10.0.2.15	DNS	121	Standard query respons
12	10.048464554	10.0.2.15	34.107.243.93	TLSv1.2	93	Application Data
13	10.052242845	34.107.243.93	10.0.2.15	TCP	60	443 → 58524 [ACK] Seq=
14	10.080929457	34.107.243.93	10.0.2.15	TLSv1.2	93	Application Data
15	10.080963869	10.0.2.15	34.107.243.93	TCP	54	58524 → 443 [ACK] Seq=
16	10.921857096	10.0.2.15	192.168.92.135	TCP	74	33098 → 80 [SYN] Seq=0
17	10.925951282	192.168.92.135	10.0.2.15	TCP	60	80 → 33098 [SYN, ACK]

Frame 17: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
Ethernet II, Src: 52:54:00:12:35:02 (52:54:00:12:35:02), Dst: 02:00:00:00:00:00
Internet Protocol Version 4, Src: 192.168.92.135, Dst: 10.0.2.15
Transmission Control Protocol, Src Port: 80, Dst Port: 33098
Source Port: 80
Destination Port: 33098
[Stream index: 3]
[Conversation completeness: Complete, WITH_DATA]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)

Can you identify the handshake process or other significant events in the captured packets?

Yes, the three-way handshake was observed between the client and the server (SYN, ACK and SYN, ACK). Which shows that combination is established.

INT303: Networking Fundamentals – Lab 2 Lab 2: Exploring Network Interfaces and Packet Transmission Using OWASP Broken Web Application IP

Exercise 1: Viewing and Configuring Network Interfaces

Task: View your network interfaces using the ifconfig or ip command. Command: ifconfig or ip addr show

Question: What network interfaces are available on your system?

The network interface is eth0

```
(Engr.ABD@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::29f2:6c5c:fdb9:5c98 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:d2:48:cb txqueuelen 1000 (Ethernet)
    RX packets 59816 bytes 54124283 (51.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34173 bytes 4848922 (4.6 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1737 bytes 138332 (135.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1737 bytes 138332 (135.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

For details about the known vulnerabilities in these applications, see
(Engr.ABD@kali)-[~]
$
```

Identify the IP address assigned to each interface.

My IP address is 10.0.2.15/24

```

(Engr.ABD@kali)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d2:48:cb brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 78543sec preferred_lft 78543sec
    inet6 fe80::29f2:6c5c:fdb9:5c98/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(Engr.ABD@kali)-[~]
$

```

Describe the difference between a loopback interface and an external network interface.

A loopback interface is a virtual network interface that routes data back to the source without using external hardware or a physical connection. It is always operational, meaning it cannot fail like traditional interfaces. While an external network interface, on the other hand, is a physical interface that connects a device to an external network, allowing communication with other devices over that network. Unlike the loopback interface, external interfaces can fail or become inactive due to hardware issues or network problems.

Exercise 2: Capturing Packets on a Specific Interface

Task: Use tcpdump or tshark to capture packets on your primary network interface. Command: tcpdump -i (e.g., tcpdump -i eth0) or tshark -i

Question:

What kind of packets are being captured?

Packets Captured are TCP, DNS, HTTP and TLSv1.2

```

File Actions Edit View Help
(Engr.ABD@kali)-[~]
$ sudo tcpdump -i eth0
[sudo] password for Engr.ABD:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:34:47.980780 IP 10.0.2.15.34196 > 192.168.92.135.http: Flags [S], seq 1437538007, win 32120, options [mss 1460,sackOK,TS val 689806487 ecr 0,nop,wscale 7], length 0
09:34:47.989994 IP 192.168.92.135.http > 10.0.2.15.34196: Flags [S.], seq 124488705, ack 1437538008, win 65535, options [mss 1460], length 0
09:34:47.990727 IP 10.0.2.15.34196 > 192.168.92.135.http: Flags [.], ack 1, win 32120, length 0
09:34:48.005354 IP 10.0.2.15.34196 > 192.168.92.135.http: Flags [P.], seq 1:510, ack 1, win 32120, length 509: HTTP: GET / HTTP/1.1
09:34:48.011384 IP 192.168.92.135.http > 10.0.2.15.34196: Flags [.], ack 510, win 65535, length 0
09:34:48.021052 IP 192.168.92.135.http > 10.0.2.15.34196: Flags [P.], seq 1:399, ack 510, win 65535, length 398: HTTP: HTTP/1.1 304 Not Modified
09:34:48.026386 IP 10.0.2.15.34196 > 192.168.92.135.http: Flags [F.], ack 399, win 31722, length 0
09:34:48.068312 IP 10.0.2.15.60604 > 192.168.92.168.domain: 9246+ PTR? 135.92.168.192.in-addr.arpa. (45)
09:34:48.134976 IP 192.168.92.168.domain > 10.0.2.15.60604: 9246 NXDomain* 0/1/0 (110)
09:34:48.175039 IP 10.0.2.15.39948 > 192.168.92.168.domain: 63885+ PTR? 15.2.0.10.in-addr.arpa. (40)
09:34:48.215395 IP 192.168.92.168.domain > 10.0.2.15.39948: 63885 NXDomain* 0/1/0 (105)
09:34:48.216215 IP 10.0.2.15.58237 > 192.168.92.168.domain: 59385+ PTR? 168.92.168.192.in-addr.arpa. (45)
09:34:48.247208 IP 192.168.92.168.domain > 10.0.2.15.58237: 59385 NXDomain* 0/1/0 (110)
09:34:53.181938 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
09:34:53.185753 ARP, Reply 10.0.2.2 is-at 52:54:00:12:35:02 (oui Unknown), length 46
09:34:53.187723 IP 10.0.2.15.59217 > 192.168.92.168.domain: 35302+ PTR? 2.2.0.10.in-addr.arpa. (39)
09:34:53.197151 IP 192.168.92.168.domain > 10.0.2.15.59217: 35302 NXDomain 0/0/0 (39)
09:34:58.041667 IP 10.0.2.15.34196 > 192.168.92.135.http: Flags [F.], ack 399, win 31722, length 0
09:34:58.042032 IP 192.168.92.135.http > 10.0.2.15.34196: Flags [F.], ack 510, win 65535, length 0
09:35:03.038385 IP 192.168.92.135.http > 10.0.2.15.34196: Flags [F.], seq 399, ack 510, win 65535, length 0
09:35:03.038751 IP 10.0.2.15.34196 > 192.168.92.135.http: Flags [F.], seq 510, ack 400, win 31722, length 0
09:35:03.039580 IP 192.168.92.135.http > 10.0.2.15.34196: Flags [F.], ack 511, win 65535, length 0
^C
22 packets captured
22 packets received by filter
0 packets dropped by kernel

(Engr.ABD@kali)-[~]
$

```

Are there any packets related to communication with the OWASP Broken Web Application VM?

Yes, this was observed through the source IP address and Handshake

```
(Engr.ABD@kali)-[~]
$ tshark -i eth0
Capturing on 'eth0'
1 0.000000000 10.0.2.15 → 192.168.92.135 TCP 54 48624 → 80 [ACK] Seq=1 Ack=1 Win=31722 Len=0
2 0.001793953 192.168.92.135 → 10.0.2.15 TCP 60 [TCP ACKed unseen segment] 80 → 48624 [ACK] Seq=1 Ack=2 Win=65535 Len=0
3 2.303427049 10.0.2.15 → 151.101.129.91 TCP 54 56502 → 443 [ACK] Seq=1 Ack=1 Win=32120 Len=0
4 2.304163980 151.101.129.91 → 10.0.2.15 TCP 60 [TCP ACKed unseen segment] 443 → 56502 [ACK] Seq=1 Ack=2 Win=65535 Len=0
5 2.692079313 151.101.129.91 → 10.0.2.15 TCP 60 443 → 56502 [RST, ACK] Seq=1 Ack=2 Win=65535 Len=0
6 2.709340523 10.0.2.15 → 192.168.92.168 DNS 87 Standard query 0xe74b A services.addons.mozilla.org
7 2.716159234 192.168.92.168 → 10.0.2.15 DNS 151 Standard query response 0xe74b A services.addons.mozilla.org A 151.101.65.91 A 151.101.1.91 A 151.101.193.91 A 151.101.129.91
8 2.720824185 10.0.2.15 → 192.168.92.168 DNS 87 Standard query 0x144f AAAA services.addons.mozilla.org
9 2.759559811 192.168.92.168 → 10.0.2.15 DNS 199 Standard query response 0x144f AAAA services.addons.mozilla.org AAAA 2a04:4e42:200::347 AAAA 2a04:4e42:400::347
AAAA 2a04:4e42::347 AAAA 2a04:4e42:600::347
10 2.782709781 10.0.2.15 → 151.101.65.91 TCP 74 46026 → 443 [SYN] Seq=0 Win=32120 Len=0 MSS=1460 SACK_PERM TSval=841579082 TSecr=0 WS=128
11 2.921263066 151.101.65.91 → 10.0.2.15 TCP 60 443 → 46026 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
12 2.921526668 10.0.2.15 → 151.101.65.91 TCP 54 46026 → 443 [ACK] Seq=1 Ack=1 Win=32120 Len=0
13 2.924883207 10.0.2.15 → 151.101.65.91 TLSv1 262 Client Hello (SNI=services.addons.mozilla.org)
14 2.926058816 151.101.65.91 → 10.0.2.15 TCP 60 443 → 46026 [ACK] Seq=1 Ack=209 Win=65535 Len=0
15 3.835876351 10.0.2.15 → 192.168.92.135 HTTP 507 [TCP Previous segment not captured] GET /dvwa HTTP/1.1
16 3.837377377 192.168.92.135 → 10.0.2.15 TCP 60 80 → 48624 [ACK] Seq=1 Ack=455 Win=65535 Len=0
17 3.900688803 192.168.92.135 → 10.0.2.15 HTTP 746 HTTP/1.1 301 Moved Permanently (text/html)
18 3.900733978 10.0.2.15 → 192.168.92.135 TCP 54 48624 → 80 [ACK] Seq=455 Ack=693 Win=31722 Len=0
19 3.935533564 10.0.2.15 → 192.168.92.135 HTTP 508 GET /dvwa/ HTTP/1.1
20 3.940360144 192.168.92.135 → 10.0.2.15 TCP 60 80 → 48624 [ACK] Seq=693 Ack=909 Win=65535 Len=0
21 4.288579819 192.168.92.135 → 10.0.2.15 HTTP 778 HTTP/1.1 302 Found
22 4.332874533 10.0.2.15 → 192.168.92.135 TCP 54 48624 → 80 [ACK] Seq=909 Ack=1417 Win=31722 Len=0
23 4.579806511 10.0.2.15 → 192.168.92.135 HTTP 569 GET /dvwa/login.php HTTP/1.1
24 4.581800196 192.168.92.135 → 10.0.2.15 TCP 60 80 → 48624 [ACK] Seq=1417 Ack=1424 Win=65535 Len=0
25 4.760641381 192.168.92.135 → 10.0.2.15 HTTP 1228 HTTP/1.1 200 OK (text/html)
26 4.760682511 10.0.2.15 → 192.168.92.135 TCP 54 48624 → 80 [ACK] Seq=1424 Ack=2591 Win=31722 Len=0
27 6.153735208 10.0.2.15 → 192.168.92.135 HTTP 495 GET /dvwa/dvwa/css/login.css HTTP/1.1
28 6.155736462 192.168.92.135 → 10.0.2.15 TCP 60 80 → 48624 [ACK] Seq=2591 Ack=1865 Win=65535 Len=0
```

Describe the role of this network interface in transmitting and receiving packets

The interface eth0 is the network connection of the host system or application, it helps in established connection between device and server.

Exercise 3: Examining Network Statistics

Task: Use the netstat or ss command to view current network statistics and connections.

Command: netstat -i or ss -s

Question:

What is the current status of your network interfaces?

The provided Kernel Interface table shows information about network interfaces on a Linux system. The key metrics include MTU (Maximum Transmission Unit), RX (receive), and TX (transmit) counters which indicate packets received and transmitted successfully or with errors. In this case, **eth0 shows 60228** received packet with no errors or drops, and **34625 transmitted packets** with no errors or drops, suggesting that the interface is functioning well. **lo**, the loopback interface has similar characteristics, indicating normal operations. The absence of errors could indicate that the firewall is properly tuned, or the hardware isn't under significant pressure.

```
(Engr.ABD@kali)-[~]
$ netstat -i
Kernel Interface table
Iface      MTU      RX-OK RX-ERR RX-DRP RX-OVR    TX-OK TX-ERR TX-DRP TX-OVR Flg
eth0       1500    60228   0     0  0      34625   0     0   0 BMRU
lo         65536   1757    0     0  0       1757   0     0   0 LRU

(Engr.ABD@kali)-[~]
$ ss -s
Total: 613
TCP: 0 (estab 0, closed 0, orphaned 0, timewait 0)

Transport Total      IP      IPv6
RAW        1         0        1
UDP         1         1         0
TCP         0         0         0
INET        2         1         1
FRAG        0         0         0

(Engr.ABD@kali)-[~]
$
```

What active connections are visible?

Using netstat -i, the kernel interface table, indicates the status of two network interfaces on a Linux system: eth0 and lo. However, this table itself does not directly show active connections. Moreover, we can summarize it as:

- Eth0:
 1. MTU (Maximum Transmission Unit)
 2. RX packets (received) and errors
 3. TX packets (transmitted) and errors
 4. Drops (discarded packets)
 5. Flags (interface status: up, down, running, etc.)

For ss -s, two active connections were found which are

- **TCP connections:** 2 (both established)
- **UDP connections:** 1

Explain the significance of these statistics in monitoring network performance

because they help network administrators:

- **Prioritize traffic:** Monitor statistics to prioritize critical traffic and ensure high-priority services get the resources they need
- **Manage bandwidth:** Use statistics to manage bandwidth allocation
- **Enforce QoS policies:** Use statistics to measure and enforce QoS policies

Also help organizations quickly identify and respond to potential security threats.

Exercise 4: Monitoring Network Traffic with Wireshark

Task: Use Wireshark to monitor network traffic on your primary interface and filter traffic going to and from the OWASP Broken Web Application VM. Command: Open Wireshark, choose your network interface, and set the filter to ip.addr == (replace with the actual IP, e.g., 192.168.56.101)

Question:

Analyze the packets going to and from the OWASP VM. What types of protocols are in use? TCP

Can you identify any key packet details such as source/destination IP addresses, port numbers, and flags?

YES, source IP address is 192.168.92.135 and Destination IP address is 10.0.2.15 and port number are 80 and 33566 respectively, as shows in the second picture the TCP flags in that Acknowledgment bit is **set** which is 1 while others are 0s

The image shows a Wireshark packet capture interface. The top pane displays a list of captured packets. The second pane shows the details of the selected packet (No. 33), which is a TCP segment from 192.168.92.135 to 10.0.2.15. The third pane shows the raw data of the packet, which is a TCP segment with flags FIN and ACK.

No.	Time	Source	Destination	Protocol	Length	Info
26	21.210327084	10.0.2.15	192.168.92.135	TCP	54	33566 → 80 [ACK] Seq=970 Ack=1900 Win=31395 Len=0
27	22.227524069	10.0.2.15	192.168.92.135	HTTP	495	GET /dwa/dwa/css/login.css HTTP/1.1
28	22.230100870	192.168.92.135	10.0.2.15	TCP	60	80 → 33566 [ACK] Seq=1900 Ack=1411 Win=65535 Len=0
29	22.234489156	10.0.2.15	192.168.92.135	TCP	54	33566 → 80 [FIN, ACK] Seq=1411 Ack=1900 Win=31395 Len=0
30	22.237885253	192.168.92.135	10.0.2.15	TCP	60	80 → 33566 [ACK] Seq=1900 Ack=1412 Win=65535 Len=0
31	22.244292726	192.168.92.135	10.0.2.15	HTTP	911	HTTP/1.1 200 OK (text/css)
32	22.244203451	192.168.92.135	10.0.2.15	TCP	60	80 → 33566 [FIN, ACK] Seq=2757 Ack=1412 Win=65535 Len=0
33	22.268666158	10.0.2.15	192.168.92.135	TCP	54	33566 → 80 [RST] Seq=1412 Win=0 Len=0
38	42.932671628	10.0.2.15	192.168.92.135	TCP	74	41916 → 80 [SYN] Seq=0 Win=32120 Len=0 MSS=1460 SACK_PERM TSval=...
39	42.935760595	192.168.92.135	10.0.2.15	TCP	60	80 → 41916 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
40	42.935807763	10.0.2.15	192.168.92.135	TCP	54	41916 → 80 [ACK] Seq=1 Ack=1 Win=32120 Len=0
41	42.942380911	10.0.2.15	192.168.92.135	HTTP	730	POST /dwa/login.php HTTP/1.1 (application/x-www-form-urlencoded...

Transmission Control Protocol, Src Port: 80, Dst Port: 33566, Seq: 2757
Source Port: 80
Destination Port: 33566
[Stream index: 1]
[Conversation completeness: Complete, WITH_DATA (63)]
[TCP Segment Len: 0]
Sequence Number: 2757 (relative sequence number)
Sequence Number (raw): 275778758
[Next Sequence Number: 2758 (relative sequence number)]
Acknowledgment Number: 1412 (relative ack number)
Acknowledgment number (raw): 2302210150
0101 = Header Length: 20 bytes (5)
Flags: 0x011 (FIN, ACK)

Flags (12 bits) (tcp.flags), 2 bytes

This image is a close-up of the TCP flags field in the packet details pane. It shows the bit fields for the flags and their corresponding values.

Acknowledgment Number: 1412 (relative ack number)
Acknowledgment number (raw): 2302210150
0101 = Header Length: 20 bytes (5)
Flags: 0x011 (FIN, ACK)
000. = Reserved: Not set
...0 = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1... = Acknowledgment: Set
.... ..0... = Push: Not set
.... ..0.. = Reset: Not set
....0. = Syn: Not set

Flags (12 bits) (tcp.flags), 2 bytes

How does the TCP/IP model apply to the data captured?

by dividing it into packets and sending it through four layers:

- Network Access Layer: The physical layer where data is transmitted between devices and the IP address is added
- Internet Layer: Sends data packets to their destination
- Host-to-Host Layer: Uses TCP to manage data flow and ensure error-free delivery
- Application Layer: Uses protocols like HTTP, HTTPS, and FTP to handle communication between user interfaces

Exercise 5: Packet Transmission Analysis

Task: Perform a ping or TCP connection request to the OWASP Broken Web Application IP and capture the packet details using tcpdump or Wireshark. Command: ping or telnet 80 (to initiate a simple connection)

Capture the traffic using tcpdump -i <interface> host <OWASP IP> or set the appropriate filter in Wireshark.

Question:

What do you observe during the packet transmission process?

It was observed in both tcpdump and Wireshark that packets were transmitted from source IP address 10.0.2.15 to destination IP address 192.168.92.135 with Internet control message protocol, echo (ping) request and during reply to the destination IP become source and source become destination IP address in response to the message.

```
File Actions Edit View Help
Engr.ABD@kali: ~ x Engr.ABD@kali: ~ x Engr.ABD@kali: ~ x
(Engr.ABD@kali)-[~]
$ sudo tcpdump -i eth0 host 192.168.92.135
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:13:56.576328 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 55, length 64
10:13:56.578694 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 55, length 64
10:13:57.577886 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 56, length 64
10:13:57.580296 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 56, length 64
10:13:58.580408 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 57, length 64
10:13:58.582879 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 57, length 64
10:13:59.582112 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 58, length 64
10:13:59.584371 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 58, length 64
10:14:00.583936 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 59, length 64
10:14:00.586299 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 59, length 64
10:14:01.585941 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 60, length 64
10:14:01.588553 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 60, length 64
10:14:02.588046 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 61, length 64
10:14:02.590474 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 61, length 64
10:14:03.590744 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 62, length 64
10:14:03.592858 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 62, length 64
10:14:04.592356 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 63, length 64
10:14:04.594759 IP 192.168.92.135 > 10.0.2.15: ICMP echo reply, id 8722, seq 63, length 64
10:14:05.594479 IP 10.0.2.15 > 192.168.92.135: ICMP echo request, id 8722, seq 64, length 64
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i
2	0.006834090	192.168.92.135	10.0.2.15	ICMP	98	Echo (ping) reply i
3	1.001540409	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i
4	1.004049567	192.168.92.135	10.0.2.15	ICMP	98	Echo (ping) reply i
5	2.002955663	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i
6	2.004775223	192.168.92.135	10.0.2.15	ICMP	98	Echo (ping) reply i
7	3.003315649	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i
8	3.005226149	192.168.92.135	10.0.2.15	ICMP	98	Echo (ping) reply i
9	4.005375463	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i
10	4.007996981	192.168.92.135	10.0.2.15	ICMP	98	Echo (ping) reply i
11	5.007173466	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i
12	5.009855127	192.168.92.135	10.0.2.15	ICMP	98	Echo (ping) reply i
13	6.015821584	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i
14	6.019074752	192.168.92.135	10.0.2.15	ICMP	98	Echo (ping) reply i
15	7.017981749	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i
16	7.020136648	192.168.92.135	10.0.2.15	ICMP	98	Echo (ping) reply i
17	8.021059521	10.0.2.15	192.168.92.135	ICMP	98	Echo (ping) request i

Frame 1: 98 bytes on wire (784 bits), 98 bytes captured on interface 0	0000	52	54	00	12	35	02	08	00	27	d2	48	cb	08	00
Ethernet II, Src: PCSSystemtec_d2:48:cb (08:00:27:00:00:00), Dst: 10.0.2.15	0010	00	54	cc	0c	40	00	40	01	45	5e	0a	00	02	0f
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 192.168.92.135	0020	5c	87	08	00	94	dc	41	fc	00	01	c9	3a	47	67
Internet Control Message Protocol	0030	00	00	43	b1	0e	00	00	00	00	00	10	11	12	13
Type: 8 (Echo (ping) request)	0040	16	17	18	19	1a	1b	1c	1d	1e	1f	20	21	22	23
Code: 0	0050	26	27	28	29	2a	2b	2c	2d	2e	2f	30	31	32	33
Checksum: 0x94dc [correct]	0060	36	37												
[Checksum Status: Good]															
Identifier (BE): 16892 (0x41fc)															
Identifier (LE): 64577 (0xfc41)															

Describe the handshake process or the round-trip of the packets for ping or TCP connection.

The handshake process established when ping, it echoes to OWASP IP address to check for its availability and readiness for communication, the packet travel between device and OWASP server and the server reply by send back the message through ICMP and communication is established.

Which layers of the OSI and TCP/IP models are involved

1. Physical Layer (Layer 1) - OSI
2. Data Link Layer (Layer 2) - OSI
3. Network Layer (Layer 3) - OSI
Internet Layer - TCP/IP
4. Transport Layer (Layer 4) - OSI
Transport Layer - TCP/IP
5. Session Layer (Layer 5) - OSI
6. Presentation Layer (Layer 6) - OSI
7. Application Layer (Layer 7) - OSI
Application Layer - TCP/IP
8. Network Access Layer - TCP/IP

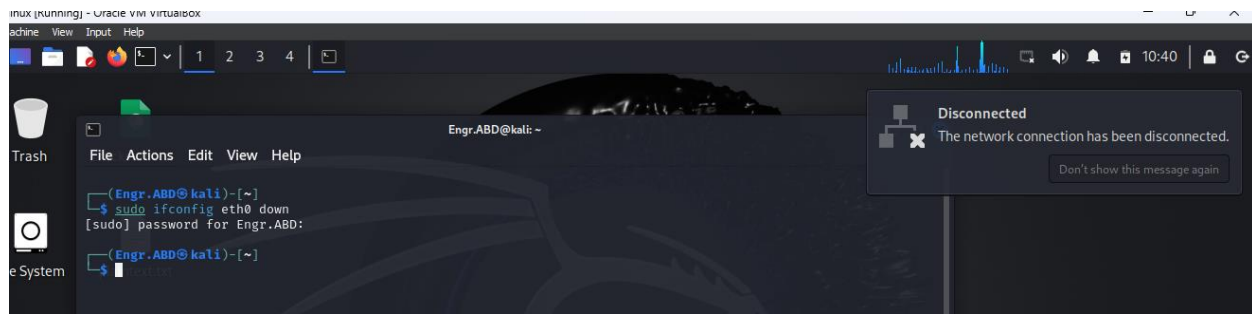
Exercise 6: Troubleshooting Network Interface Issues

Task: Simulate a network interface failure by disabling and then re-enabling an interface. Observe how this affects network connectivity to the OWASP Broken Web Application VM.

Command: `sudo ifconfig down` and `sudo ifconfig up` (or use `ip` commands)

Question: What happens when you disable the network interface?

The network connection has been disconnected as shown in the diagram below



How does your system respond when the interface is re-enabled?

The system refreshes its connection to the network and the internet is back

Explain how network administrators can use this knowledge for troubleshooting connectivity issues.

This process helps in resetting the interface and can assist in diagnosing connectivity problems.

Exercise 7: Bandwidth Monitoring

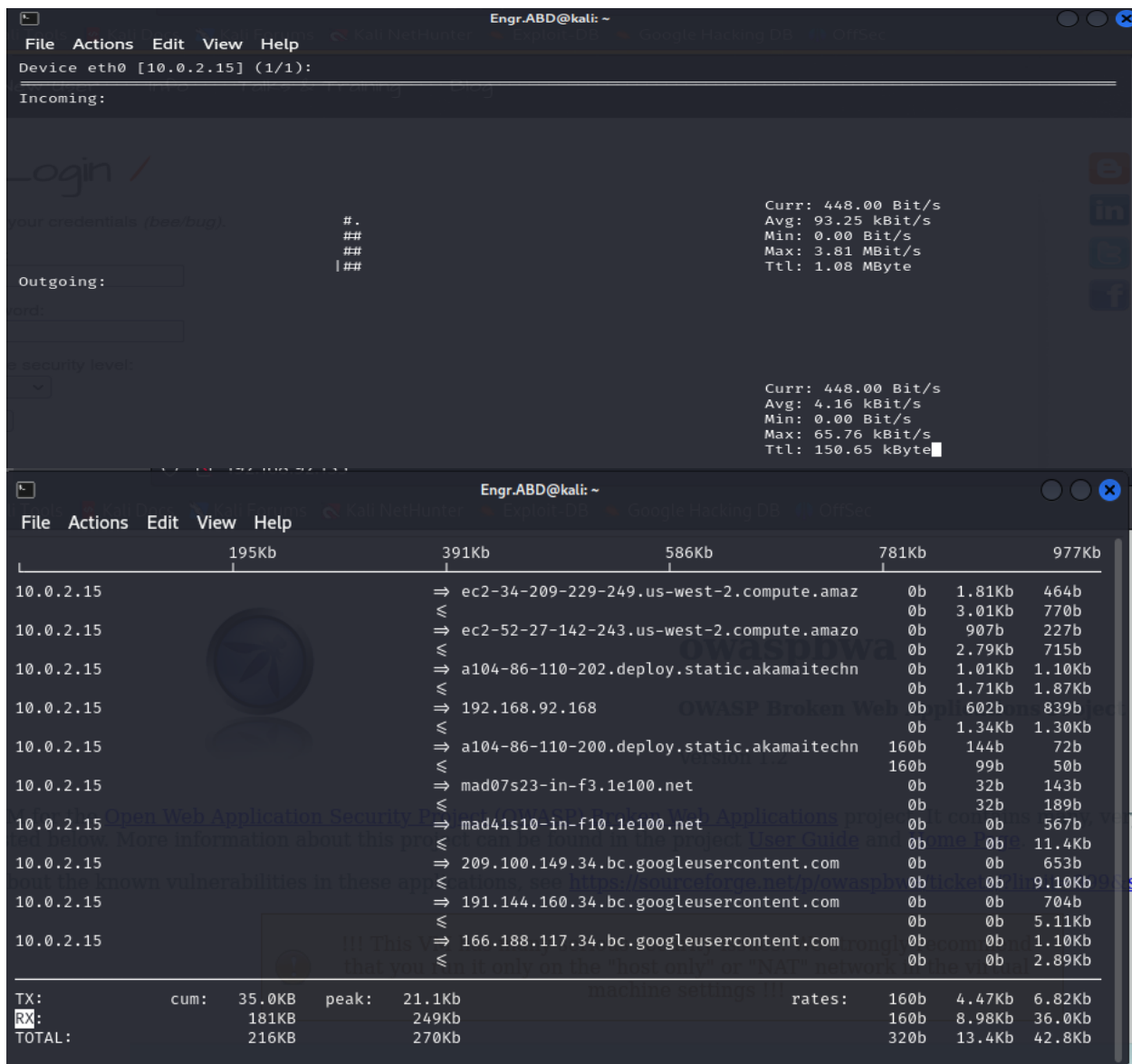
Task: Use the `iftop <interface>` or `nload <interface>` command to monitor the bandwidth usage on your system while interacting with the OWASP Broken Web Application VM.

Command: `sudo iftop -i <interface>` or `nload <interface>`

Question:

What is the current bandwidth usage while communicating with the OWASP VM?

Both `iftop` and `nload` are used for monitoring bandwidth as shown in the pictures below. The current bandwidth usage while communicating with the OWASP is 448.00 bit/sec. This can be observed in both incoming and outgoing traffic using **nload**, on `iftop` it is 320 bit/sec. The difference observed between the two (`nload` and `iftop`) is as a result of different engagement with the OWASP web browser.



Identify the impact of network traffic on your interface. Is there any traffic congestion?

No congestion. This also shows the incoming and outgoing traffic for monitoring, iftop capture and categories three (3) different period of bandwidth and these were classified in 2sec/bit, 10sec/kb and 40sec/kb. Although the bandwidth can be Mb.

How does this help in monitoring

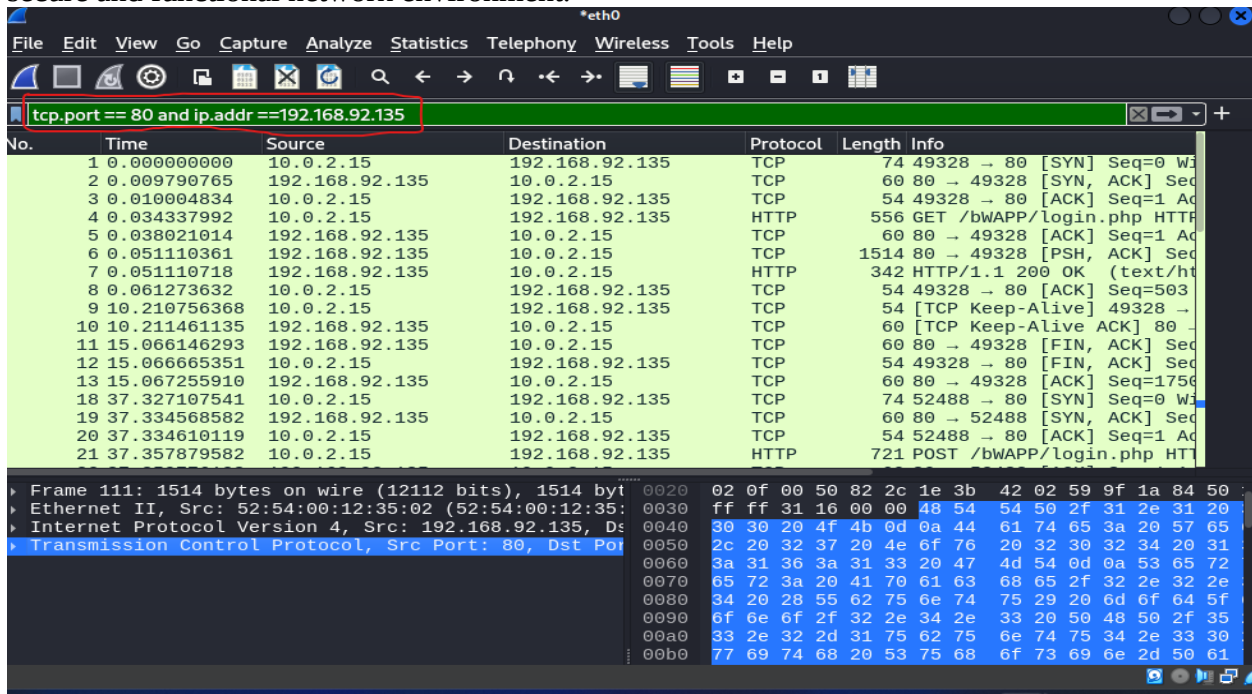
This will help to monitor network traffic to know whether is flooded or not and it help monitor an attacks such as DDos attack. And it is essential to monitor it time to time inorder to check if anything is wrong in the network

Exercise 8: Advanced Packet Capture Filters

Task: Create custom filters in Wireshark or tcpdump to capture only specific types of traffic, such as TCP or HTTP traffic, between your system and the OWASP VM. Example Filter for HTTP Traffic: tcp port 80 and ip.addr ==

Question: What is the significance of filtering specific traffic?

Filtering specific traffic is significant for several reasons, including enhancing network security, optimizing performance, and managing bandwidth. By blocking unwanted packets or traffic from specific sources, administrators can reduce the risk of malware and attacks, ensure that essential services are prioritized, and improve overall network efficiency. Allow you to focus on analysis specific traffic very well for necessary actions. This method is crucial for maintaining a secure and functional network environment.



How can advanced filters help network engineers diagnose and resolve issues?

Advanced filters can help network engineers diagnose and resolve issues by allowing them to isolate and analyze specific traffic patterns or anomalies within the network. This capability enables engineers to focus on relevant data, identify problematic protocols, and pinpoint the root causes of connectivity issues. By utilizing advanced filtering tools, such as Wireshark or tcpdump, engineers can effectively narrow down the scope of their troubleshooting efforts, making the diagnosis process more efficient and effective.

INT303: Networking Fundamentals – Lab 3 Lab 3: TCP/IP Protocol Stack and Packet Inspection Using OWASP Broken Web Application IP

Lab Exercises:

Exercise 1: Understanding the TCP/IP Model

Task: Briefly review the layers of the TCP/IP model (Application, Transport, Network, and Link layers). Understand the purpose and function of each layer in data transmission.

Question: Explain the differences between the TCP/IP and OSI models.

S/N	Characteristics	TCP/IP Model	OSI Model
1	Number of Layers	4	7
2	Layering Structure	Combined layers	Separate layers
3	Focus	Communication protocols	Network architecture
4	Layers	Application, Transport, Internet, Network Access.	Application, Presentation, Session, Transport, Network, Data Link, Physical.

Which layers of the TCP/IP model correspond to specific OSI layers?

- **Application**

The TCP/IP Application layer corresponds to the OSI Application, Session, and Presentation layers. The Application layer handles process-to-process communication functions.

- **Transport**

The TCP/IP Transport layer corresponds to the OSI Transport layer.

- **Internet**

The TCP/IP Internet layer corresponds to the OSI Network layer. The Internet layer accepts and delivers packets for the network.

- **Data link**

The TCP/IP Data link layer corresponds to the OSI Data link layer. The data link layer handles packaging and unpacking data in frames.

Exercise 2: Capturing and Analyzing TCP Packets

Task: Initiate a connection to the OWASP Broken Web Application VM via HTTP or SSH.

Capture TCP packets related to this connection using tcpdump or Wireshark.

Command: tcpdump -i tcp and host or set a filter in Wireshark. Example: tcpdump -i eth0 tcp and host 192.168.56.101

Question: What happened during the TCP handshake (SYN, SYN-ACK, ACK)?

During the TCP three-way handshake, the client with IP address 10.0.2.15 initiated the connection by sending a SYN packet to the server at 192.168.92.135. In response, the server acknowledged the client's SYN packet with a SYN-ACK packet, which also included its own SYN request. The client then completed the handshake by sending an ACK packet back to the server, acknowledging the server's SYN-ACK. This exchange successfully establishes a TCP connection between the client and the server. The two pictures below show the handshake capture using Wireshark and tcpdump.

Time	Source	Destination	Protocol	Length	Info
3 3.621177529	10.0.2.15	192.168.92.135	TCP	74	38444 → 80 [SYN] Seq=0 Win=32120
4 3.624224273	192.168.92.135	10.0.2.15	TCP	60	80 → 38444 [SYN, ACK] Seq=0 Ack=
5 3.624571896	10.0.2.15	192.168.92.135	TCP	54	38444 → 80 [ACK] Seq=1 Ack=1 Win
6 3.632579015	10.0.2.15	192.168.92.135	HTTP	601	GET / HTTP/1.1
7 3.636342570	192.168.92.135	10.0.2.15	TCP	60	80 → 38444 [ACK] Seq=1 Ack=548 W
8 3.640953749	192.168.92.135	10.0.2.15	HTTP	452	HTTP/1.1 304 Not Modified
9 3.661504396	10.0.2.15	192.168.92.135	TCP	54	38444 → 80 [ACK] Seq=548 Ack=399
0 13.667783078	10.0.2.15	192.168.92.135	TCP	54	[TCP Keep-Alive] 38444 → 80 [ACK
1 13.670222314	192.168.92.135	10.0.2.15	TCP	60	[TCP Keep-Alive ACK] 80 → 38444
2 18.655897887	192.168.92.135	10.0.2.15	TCP	60	80 → 38444 [FIN, ACK] Seq=399 Ac
3 18.656438397	10.0.2.15	192.168.92.135	TCP	54	38444 → 80 [FIN, ACK] Seq=548 Ac
4 18.664443541	192.168.92.135	10.0.2.15	TCP	60	80 → 38444 [ACK] Seq=400 Ack=549
9 27.476817581	10.0.2.15	172.67.10.39	TCP	74	38424 → 80 [SYN] Seq=0 Win=32120
0 27.617101718	172.67.10.39	10.0.2.15	TCP	60	80 → 38424 [SYN, ACK] Seq=0 Ack=
1 27.617779615	10.0.2.15	172.67.10.39	TCP	54	38424 → 80 [ACK] Seq=1 Ack=1 Win
6 28.682427883	10.0.2.15	104.18.13.149	TCP	74	45000 → 443 [SYN] Seq=0 Win=3212
1 28.785579151	10.0.2.15	64.225.91.73	TCP	74	60258 → 80 [SYN] Seq=0 Win=32120

```

File Actions Edit View Help
Engr.ABD@kali: ~ x Engr.ABD@kali: ~ x
(Engr.ABD@kali)~[~]
$ sudo tcpdump -i eth0 tcp and host 192.168.92.135

[sudo] password for Engr.ABD:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:32:46.731265 IP 10.0.2.15.40642 > 192.168.92.135.http: Flags [S], seq 2730737764, win 32120, options [mss 146
0,sackOK,TS val 2234202304 ecr 0,nop,wscale 7], length 0
11:32:46.736354 IP 192.168.92.135.http > 10.0.2.15.40642: Flags [S.], seq 760448001, ack 2730737765, win 65535,
options [mss 1460], length 0
11:32:46.737561 IP 10.0.2.15.40642 > 192.168.92.135.http: Flags [.], ack 1, win 32120, length 0
11:32:46.763342 IP 10.0.2.15.40642 > 192.168.92.135.http: Flags [P.], seq 1:668, ack 1, win 32120, length 667: H
TTP: POST /bWAPP/login.php HTTP/1.1
11:32:46.765692 IP 192.168.92.135.http > 10.0.2.15.40642: Flags [.], ack 668, win 65535, length 0
11:32:46.788048 IP 192.168.92.135.http > 10.0.2.15.40642: Flags [P.], seq 1:1461, ack 668, win 65535, length 146
0: HTTP: HTTP/1.1 200 OK
11:32:46.788048 IP 192.168.92.135.http > 10.0.2.15.40642: Flags [P.], seq 1461:1782, ack 668, win 65535, length
321: HTTP
11:32:46.798384 IP 10.0.2.15.40642 > 192.168.92.135.http: Flags [.], ack 1782, win 30660, length 0
11:32:56.976023 IP 10.0.2.15.40642 > 192.168.92.135.http: Flags [.], ack 1782, win 30660, length 0
11:32:56.979871 IP 192.168.92.135.http > 10.0.2.15.40642: Flags [.], ack 668, win 65535, length 0
11:33:01.801014 IP 192.168.92.135.http > 10.0.2.15.40642: Flags [F.], seq 1782, ack 668, win 65535, length 0
11:33:01.805780 IP 10.0.2.15.40642 > 192.168.92.135.http: Flags [F.], seq 668, ack 1783, win 30660, length 0
11:33:01.809038 IP 192.168.92.135.http > 10.0.2.15.40642: Flags [.], ack 669, win 65535, length 0

```

Identify and describe the flags used in TCP communication (SYN, ACK, FIN, etc.).

During the SYN phase of the TCP three-way handshake, the SYN flag is set to 1, indicating a connection request from the client. When the server responds, it sets both the SYN and ACK flags to 1 in its response packet, acknowledging the client's SYN while also signaling its own request to establish a connection. Finally, the client acknowledges the server's response by sending a packet with the ACK flag set to 1, completing the handshake.

How does the TCP connection maintain reliability during transmission?

TCP (Transmission Control Protocol) maintains reliability during transmission through several key mechanisms:

1. **Sequence Numbers:** Each byte of data is assigned a unique sequence number, allowing the receiving end to reorder packets and detect any lost packets.
2. **Acknowledgments (ACKs):** After receiving data, the receiving device sends back an acknowledgment for each segment (or multiple segments, depending on the implementation). This informs the sender that the data was received successfully.
3. **Retransmission:** If the sender does not receive an acknowledgment for a specific packet within a certain time frame (due to packet loss, for example), it will retransmit that packet.
4. **Error Checking:** TCP includes a checksum that verifies the integrity of the data. If the checksum does not match the receiving end, the packet is discarded, and the sender will retransmit the data.
5. **Flow Control:** TCP uses flow control mechanisms, such as the sliding window protocol, to prevent the sender from overwhelming the receiver with too much data at once. This ensures that the receiver can process incoming segments at a manageable rate.
6. **Congestion Control:** TCP employs algorithms (like TCP Tahoe and TCP Reno) to monitor network congestion. If a packet loss is detected, the sender reduces the transmission rate, allowing the network to recover.
7. **Connection Establishment and Termination:** TCP establishes a reliable connection through a three-way handshake (SYN, SYN-ACK, ACK) and ensures a graceful termination of the connection to properly close any open state.

Together, these mechanisms make TCP a reliable protocol for ensuring that data is delivered accurately and in order, making it suitable for applications that require a guarantee of data integrity, such as web browsing, email, and file transfers.

Exercise 3: Investigating IP Packets (Network Layer)

Task: Capture and analyze the IP packets from your connection to the OWASP Broken Web Application.

Command: Use Wireshark or tcpdump with the filter `ip.addr ==` to capture IP packets.

Question: What fields can you see in the IP packet header (e.g., Source IP, Destination IP, TTL, etc.)?

The IPv4 packet contains key information such as the version, header length, total length, TTL, and protocol type (TCP). The header checksum is provided for error detection, although its validation is currently disabled in this capture.

What is the significance of each of these fields?

IPv4 Header Details

- **Version:** 4 - This indicates that the packet is using IPv4.
- **Header Length:** 20 bytes (5) - The length of the IPv4 header is 20 bytes, which corresponds to a value of 5 (indicating $5 * 4$ bytes).
- **Differentiated Services Field:** 0x00 - This field is used for Quality of Service (QoS) and indicates that the default service is being used (CS0).
 - **DSCP:** Default (0) - The Differentiated Services Codepoint is set to the default.
 - **ECN:** Not ECN-Capable Transport (0) - The Explicit Congestion Notification is not enabled.
- **Total Length:** 44 - This indicates the total length of the IP packet, including the header and data (payload).

- **Time to Live (TTL):** 64 - The maximum number of hops the packet can take before being discarded. A value of 64 is common for IP packets.
- **Protocol:** TCP (6) - This indicates that the transport layer protocol is TCP, which is identified by the number 6.
- **Header Checksum:** 0x4502 - This is a checksum used for error-checking the header. The validation is disabled, so the checksum status is unverified.
- **Source Address:** 192.168.92.135 - The IP address of the sender.
- **Destination Address:** 10.0.2.15 - The IP address of the recipient.

How does IP routing work in this scenario? Are there any hops between your system and the OWASP VM?

YES, the first hop initiates the communication from your device through the local area network to a router or switch. Subsequent hops involve the router making routing decisions based on its routing table, sending the packet through various routers until it reaches the OWASP VM. Each router along the way processes the packet and forwards it until it arrives at the final destination.

Exercise 4: Application Layer Analysis (HTTP/SSH Traffic)

Task: If you're using HTTP, filter the packets to capture only HTTP traffic using Wireshark or tcpdump. If you're using SSH, capture and analyze the encrypted traffic.

Command for HTTP traffic: tcpdump -i port 80 and host Command for SSH traffic: tcpdump -i port 22 and host Command

Question: Analyze the HTTP packets. What information is available in the HTTP request and response?

The http contains request method: GET, Request URI and Request version: HTTP/1.1, host: 192.168.92.135, the response in frame: 22, this packet will be responded in the packet with this number.

For SSH traffic, what is the significance of encrypted packets? Can you analyze the payload?

How does the application layer play a role in data exchange between your system and the OWASP VM?

The application layer is vital for facilitating data exchange between my system and the OWASP VM by using appropriate protocols, managing data formats, providing user interfaces, handling sessions, and ensuring security throughout the communication process. Whether through a web browser, command-line tool, or application, the application layer enables meaningful interactions with the services running on the OWASP VM.

Exercise 5: Error Handling in TCP/IP Transmission

Task: Simulate packet loss or network issues by introducing a delay or stopping the network connection momentarily. Capture how TCP/IP handles this issue. Command: Temporarily disable your network interface using ifconfig down and then re-enable it.

Question:

What happens when packets are dropped or delayed?

Disabling and then re-enabling a network interface leads to immediate packet loss during the downtime, impacting any ongoing connections. On re-enabling, connectivity is restored, but packets sent during that period are lost. TCP and UDP handle this scenario differently, with TCP providing reliable retransmission while UDP leaves packet delivery unguaranteed. Users may experience delays and disconnections in active applications as well.

How does TCP ensure data reliability in the presence of errors?

- TCP is designed for reliability and will handle dropped packets by retransmitting them. If a TCP connection is disrupted due to the interface being down, it will time out and attempt to re-establish the connection.
- TCP's congestion control will kick in if significant packet loss occurs, potentially slowing down subsequent transmissions

How do retransmissions and sequence numbers work in TCP to maintain a proper data flow?

TCP uses sequence numbers to identify and order data segments, ensuring that they are reassembled correctly at the receiver's end. Retransmissions ensure that lost packets are resent to maintain data integrity. Together with flow control and congestion control mechanisms, these features enable TCP to provide a reliable, orderly, and efficient data flow across networks.

Exercise 6: ICMP and Ping Inspection (Network Layer)

Task: Send a series of ping commands to the OWASP Broken Web Application VM and capture the ICMP packets. Command: ping and use Wireshark or tcpdump to capture ICMP packets with the filter icmp.

Question: What are the key fields in an ICMP packet (e.g., Type, Code, Checksum)?

1. The icmp packet type is 8 (Echo (ping) request), code: 0 and checksum: 0x5f68 [correct], checksum status: Good. **ICMP Packet Type: 8 (Echo Request)**
 - The type field in ICMP specifies the kind of message being sent. For ICMP, type 8 indicates an **Echo Request**. This is the type of packet sent when you perform a ping, which is used to check if a specific IP address (host) is reachable over the network.
2. **Code: 0**
 - The code field specifies further details about the type of ICMP message. For type 8 (Echo Request), the code is typically 0, indicating a standard Echo Request without any additional options or conditions.
3. **Checksum: 0x5f68**
 - The checksum is a 16-bit field used for error-checking. It helps determine if the data in the packet has been corrupted during transmission. The calculated checksum (0x5f68 in this case) indicates the integrity of the packet.
 - The phrase “**checksum status: Good**” means that the checksum has been correctly calculated, and the packet data appears to be intact.

Function of an ICMP Echo Request

- **Ping Functionality:** When a host sends an ICMP Echo Request, it typically expects an ICMP Echo Reply (type 0) from the destination. This exchange helps verify that the destination device is reachable and functioning.

- The ICMP packet you described is an Echo Request intended to test the reachability of a network device. It correctly specifies the type, code, and a valid checksum, indicating that it is functioning properly for a ping test. If this packet successfully reaches the destination, you would typically expect to receive a corresponding ICMP Echo Reply.



What is the significance of TTL in both ICMP packets and general IP packets?

1. **Prevention of Routing Loops:** It ensures that packets are not trapped in infinite routing loops.
2. **Network Resource Management:** By discarding expired packets, it maintains efficient packet handling and reduces load on routers.
3. **Troubleshooting and Diagnostics:** TTL is used in tools like ping and traceroute to analyze network paths and diagnose connectivity issues.
4. **Network Stability and Security:** It promotes network stability and can have security implications by limiting information about the network structure.

Exercise 7: Analyzing UDP Packets (Transport Layer)

Task: If available, generate some UDP traffic to the OWASP VM by using a simple application or UDP-based service. Capture and analyze the UDP packets. Command: `tcpdump -i udp and host`

Question: Compare UDP with TCP. What are the major differences in packet structure and behavior?

TCP is reliable and ensures ordered delivery, making it suitable for many applications, while UDP is fast and efficient, making it ideal for scenarios where speed is more critical than reliability.

Why does UDP not ensure reliability, and in what scenarios would you prefer UDP over TCP?

UDP does not ensure reliability because it operates on a connectionless basis, lacks error correction and retransmission mechanisms, and is designed to minimize overhead. Its characteristics make it suitable for applications that require low latency and can tolerate some degree of data loss, such as real-time communication, online gaming, and multimedia streaming. By choosing UDP for these scenarios, applications can benefit from faster transmission times at the expense of guaranteed delivery.

How does UDP manage data transmission without the need for acknowledgments or retransmissions?

UDP's design allows for rapid and efficient data transmission without acknowledgments or retransmissions by using a simple, connectionless model that emphasizes speed over reliability. It is particularly suited for applications that can tolerate some loss and prioritize low latency, such as real-time communication and multimedia streaming. Application developers can implement their own reliability mechanisms, when necessary, but the inherent simplicity and low overhead of UDP are its defining features.

Exercise 8: OS Detection via Nmap (Network and Transport Layers)

Task: Use nmap to perform OS detection on the OWASP Broken Web Application VM, analyzing how nmap identifies the operating system based on packet behavior. Command: `nmap -O`

Question:

How does nmap detect the OS based on the captured packets?

By leveraging distinct characteristics of the TCP/IP stack, analyzing various types of packet responses, and referencing its fingerprint database, Nmap can successfully identify the operating system of a target host. This capability is invaluable for network administrators and security professionals for assessing network security, identifying vulnerabilities, and managing assets. If you have further questions or need more detailed information on any aspect of Nmap's operation.

What packet characteristics (TTL, window size, etc.) help in identifying the OS?

Here's a simplified table of some of the typical values associated with different operating systems:

Table

Feature	Windows	Linux	Mac OS	FreeBSD	Solaris
---------	---------	-------	--------	---------	---------

Default TTL	128	64	255	64	255
Initial Sequence Numbers	Random or predictable patterns	Randomized	Randomized	Randomized	Randomized
Typical TCP Window Size	65535	Varies	Varies	Varies	Varies
Common TCP Options	Timestamps, Window Scale	Selective Acknowledgments	Window Scale	Timestamps	Timestamps
Common ICMP Responses	Specific format and content	Default formats	More verbose	Different structures	Different structures

By examining these packet characteristics through its fingerprinting techniques, Nmap can accurately guess the operating system of the target host. This process helps network administrators and security professionals gain useful insights into the devices on their networks.

Explain why OS detection can be an important step in network analysis and vulnerability assessment.

OS detection is a vital component of effective network analysis and vulnerability assessment. It helps organizations identify vulnerabilities, tailor security measures, manage assets, assess risks, meet compliance requirements, enhance incident response, optimize performance, and improve threat hunting capabilities. By knowing the operating systems in play, organizations can implement more robust security strategies that are aligned with their specific infrastructure needs, ultimately reducing the risk of security breaches and improving the overall security posture.

Exercise 9: Analyzing ARP Traffic (Link Layer)

Task: Use arp-scan or tcpdump to capture ARP traffic in your local network, focusing on communication with the OWASP VM. Command: arp-scan -I

Question:

What information can you gather from ARP packets (e.g., MAC addresses)?

ARP packets provide essential information that aids in understanding network topology and device connectivity. By analyzing MAC and IP addresses, along with other fields in the ARP packet, network administrators can accurately map and monitor their networks while identifying potential issues or threats.

How does the ARP protocol function at the link layer?

ARP plays a vital role in enabling devices on a local network to communicate effectively by resolving IP addresses into MAC addresses. Its operation involves broadcasting requests, receiving replies, and updating ARP caches, all of which occur at the link layer of the OSI model. By facilitating this address resolution, ARP ensures that data packets can be correctly delivered to the appropriate hardware addresses for successful local network communication.

What role does ARP play in the communication between your system and the OWASP VM?

ARP facilitates the essential link between IP addresses and MAC addresses in the communication between your system and the OWASP VM. It ensures that your device can discover the MAC address of the VM, allowing data packets to be properly transmitted over the local network.

Exercise 10: Troubleshooting Network Connectivity Using TCP/IP Knowledge

Task: Simulate a common network issue (e.g., incorrect subnet mask, gateway misconfiguration) and troubleshoot the issue using your understanding of the TCP/IP stack.

Question:

What is the issue you simulated, and how did it affect network communication?

In simulating an ARP spoofing attack, you can observe firsthand how it compromises network communication, affects data integrity and confidentiality, and impacts overall network performance. Understanding these implications is crucial for applying appropriate defenses against such attacks in real-world network environments.

How did you diagnose and resolve the issue using packet inspection and knowledge of the TCP/IP layers?

By leveraging packet inspection to analyze ARP traffic and understanding the TCP/IP architecture, the diagnosis and resolution of an ARP spoofing attack can be efficiently managed. This approach involves observation of symptoms, capturing and analyzing packets, isolating rogue devices, implementing remedial actions, and following up with enhanced security measures to prevent recurrence.

What tools and techniques would you recommend for real-world network troubleshooting?

Combining various tools with proven techniques creates a robust arsenal for troubleshooting network issues in real-world scenarios. The appropriate use of packet capture tools, command line utilities, and system monitoring applications, along with structured troubleshooting approaches, can significantly improve your effectiveness in diagnosing and resolving network problems.